

CHESTER COUNTY AP, PA, USA

WMO: 720408

Lat: 39.983N

Lon: 75.867W

Elev: 660

StdP: 14.35

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.1	14.3	-4.0	4.5	13.3	0.5	5.8	17.5	27.5	26.5	25.5	26.6	9.7	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	89.9	76.1	87.5	74.8	84.1	73.0	78.6	85.3	76.8	83.5	75.2	81.2	6.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.8	143.2	82.4	74.8	133.7	79.8	73.1	126.0	77.6	42.9	84.6	40.5	84.7	39.2	80.4	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	19.6	16.8	DB	5.5	93.4	5.2	3.9	1.7	96.2	-1.4	98.5	-4.3	100.7	-8.1	103.6	(4)
(5)			WB	3.6	81.9	4.7	3.7	0.2	84.6	-2.5	86.7	-5.1	88.8	-8.5	91.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.6	30.6	33.7	40.8	52.6	62.5	70.3	75.1	72.5	66.9	55.8	44.1	36.7		
	DBStd	17.24	9.58	9.85	9.23	8.90	7.89	5.70	4.95	4.57	6.63	8.22	8.23	8.66		
	HDD50	2126	604	465	312	69	4	0	0	0	0	35	213	424		
	HDD65	5113	1067	877	750	384	146	19	0	2	57	305	628	878		
	CDD50	3427	2	8	28	149	392	608	778	696	506	214	35	11		
	CDD65	936	0	0	0	14	68	176	313	233	113	19	0	0		
	CDH74	6799	0	0	5	128	504	1295	2578	1510	695	84	0	0		
	CDH80	1909	0	0	0	24	123	332	870	364	185	11	0	0		
	WSAvg	6.2	7.7	7.9	8.1	7.5	5.8	5.0	4.1	4.0	4.5	5.9	6.8	7.2		
	PrecAvg	46.9	3.7	2.6	4.4	3.8	3.5	4.5	4.1	4.0	4.8	4.0	3.4	3.9		
(15) Precipitation	PrecMax	64.3	6.1	5.8	7.6	7.4	7.0	11.3	7.8	7.7	12.8	10.4	5.0	8.1		
	PrecMin	34.7	1.6	0.5	0.5	1.7	1.4	1.9	1.1	1.6	0.4	0.7	1.0	0.9		
	PrecStd	8.2	1.3	1.1	1.7	1.5	1.5	2.6	1.8	1.5	3.2	2.5	1.2	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.0	67.7	72.8	83.6	88.1	90.8	93.3	90.0	89.7	81.1	69.7	64.0		
		MCWB	56.5	56.8	59.7	63.3	73.6	76.2	77.7	76.1	75.1	69.3	60.0	61.9		
	2%	DB	54.3	59.1	65.7	78.6	83.6	87.6	90.4	86.2	85.6	75.4	64.2	59.1		
		MCWB	51.2	52.3	56.9	63.5	72.1	74.0	76.7	74.9	72.5	67.7	57.8	57.5		
	5%	DB	48.2	53.8	60.9	72.5	80.7	83.6	87.9	83.8	81.4	72.2	61.0	54.6		
		MCWB	45.2	47.3	52.4	61.6	69.1	72.6	75.7	73.7	70.6	66.1	55.3	52.6		
	10%	DB	44.7	46.4	55.1	67.9	75.3	81.3	84.3	81.6	77.5	68.3	56.8	49.7		
		MCWB	41.3	41.5	48.7	58.2	66.1	70.4	73.9	71.9	68.2	63.4	51.5	47.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.4	59.1	62.4	72.0	78.5	80.8	82.5	80.1	77.3	73.2	64.2	63.2		
		MCDB	57.5	64.4	68.7	75.2	83.6	83.9	85.7	84.9	85.3	75.6	65.4	63.4		
	2%	WB	52.1	52.9	58.2	66.1	73.3	76.4	79.3	77.6	75.1	70.3	60.1	57.5		
		MCDB	53.6	56.7	63.4	73.1	78.8	83.6	86.5	84.0	81.6	74.2	62.9	59.0		
	5%	WB	45.8	46.1	54.2	62.7	71.3	74.4	77.2	75.5	73.3	67.6	56.7	53.8		
		MCDB	47.4	50.0	59.1	70.5	77.9	80.8	84.4	80.3	77.8	71.0	59.7	54.8		
	10%	WB	41.1	42.4	49.9	59.5	68.9	72.6	75.5	73.9	71.7	64.0	53.6	47.1		
		MCDB	44.1	46.4	55.3	66.5	74.0	78.5	82.0	78.5	76.0	66.5	56.4	48.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.6	15.6	18.3	17.3	17.0	16.6	15.9	16.3	15.7	14.7	11.9		
		MCDBR	19.2	20.9	23.0	23.3	19.7	19.5	18.6	17.2	18.7	18.0	18.9	16.6		
		MCWBR	17.2	16.4	15.8	12.9	10.9	9.4	8.8	7.9	8.8	10.5	13.3	15.8		
	5% WB	MCDBR	17.8	19.0	21.7	21.5	18.3	17.2	16.5	14.9	16.0	14.9	16.2	16.3		
		MCWBR	17.2	16.3	17.4	14.2	11.1	9.9	9.0	8.2	8.9	10.6	14.4	16.7		
(40) Clear-Sky Solar Irradiance	taub		0.305	0.316	0.351	0.389	0.430	0.493	0.496	0.495	0.409	0.355	0.319	0.302		
	taud		2.455	2.404	2.379	2.293	2.212	2.074	2.081	2.069	2.281	2.446	2.510	2.504		
	Ebn at Noon		274	286	286	281	270	252	250	246	263	268	266	266		
	Edh at Noon		25	30	35	41	45	52	51	50	38	29	24	22		
(44) All-Sky Solar Radiation	RadAvg		602	886	1172	1519	1699	1867	1892	1653	1361	974	691	520		
	RadStd		52	103	90	134	205	148	92	100	163	105	62	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+131

Nomenclature: See separate page